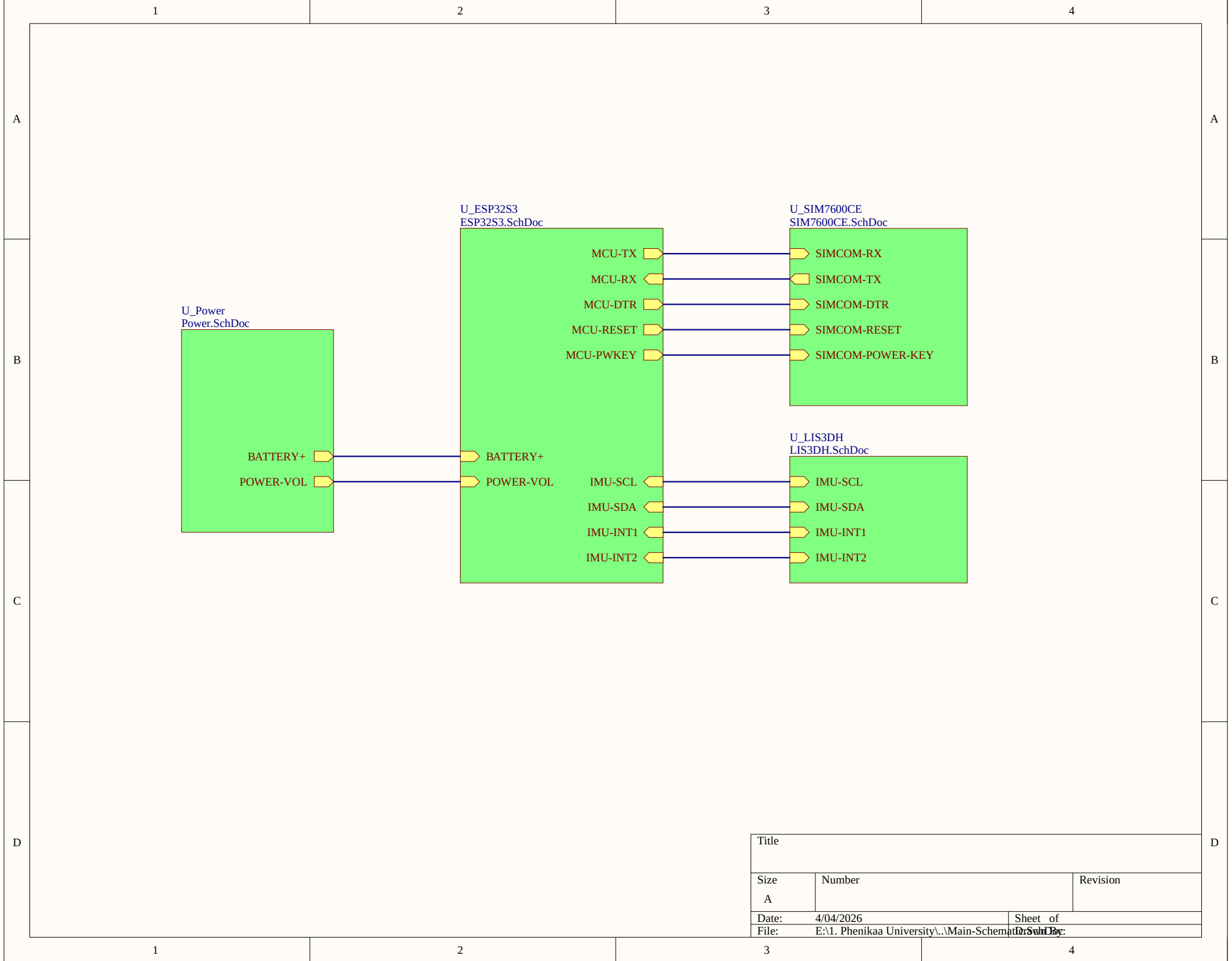
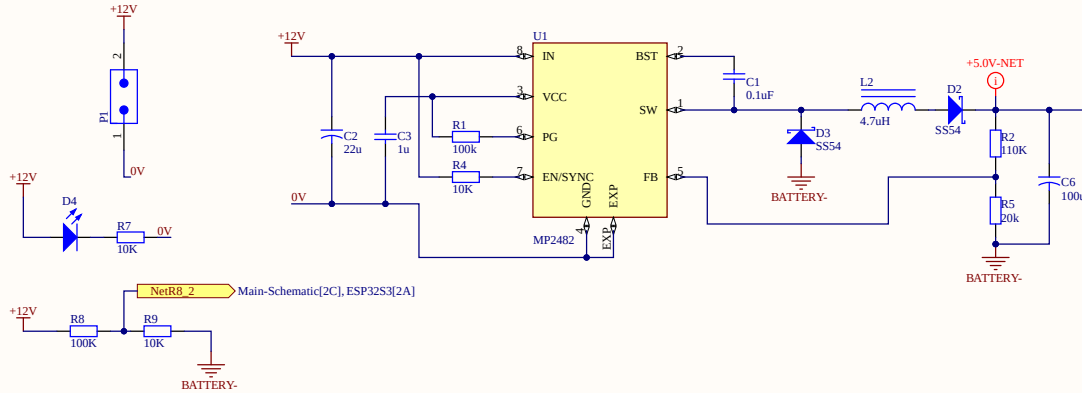


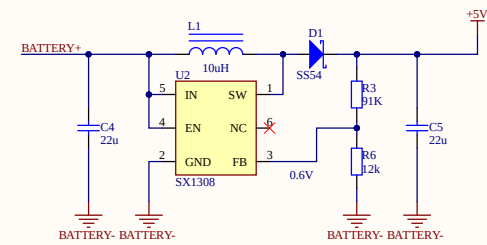


Title		
Size A	Number	Revision
Date: 4/04/2026	Sheet of	
File: E:\1. Phenikaa University\..\Main-Schematic\U_Power.SchDoc	U_Power.SchDoc	

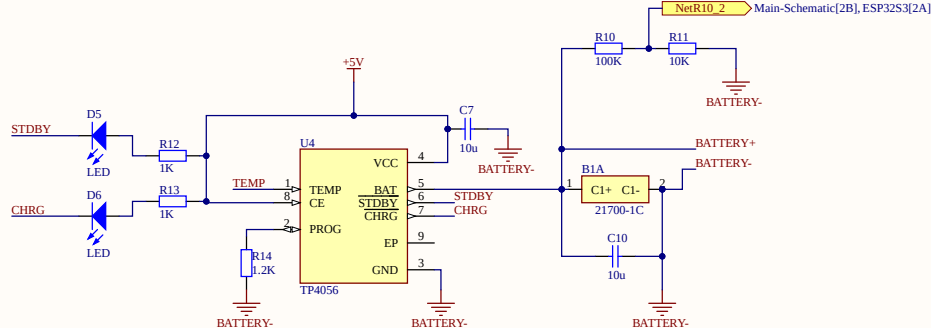
12-24V -> 5V



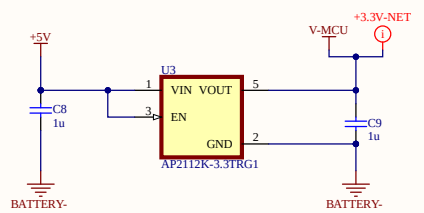
5V BOOST



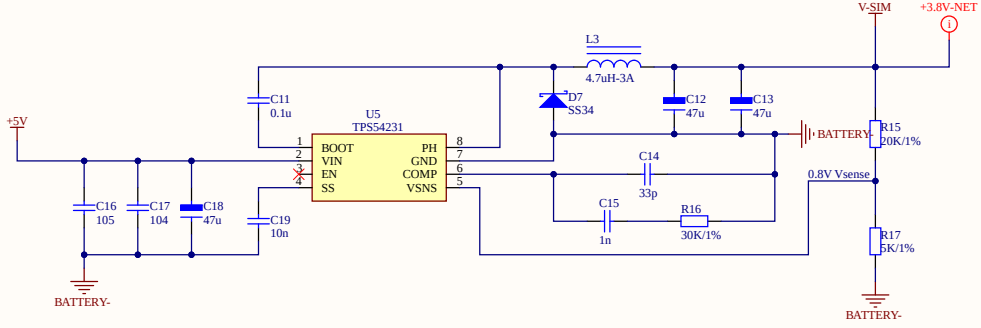
1S LI-ION CHARGER

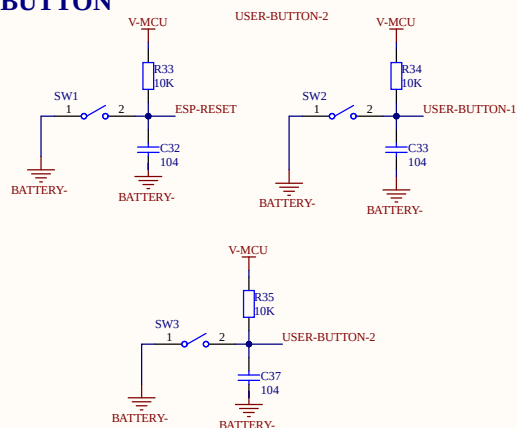
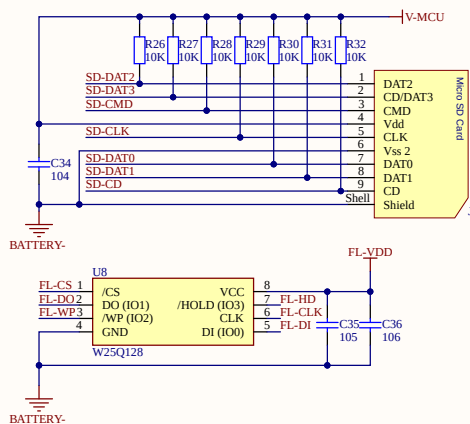
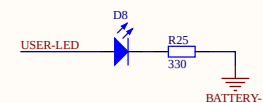
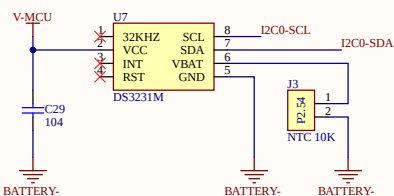
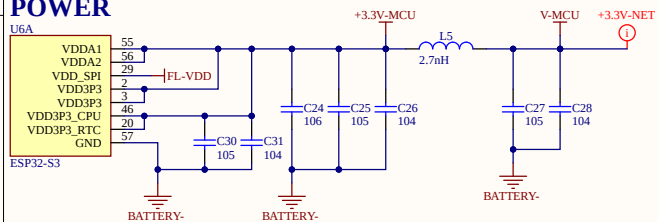
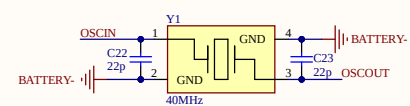
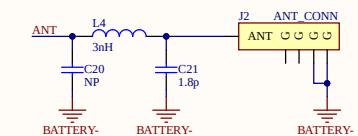
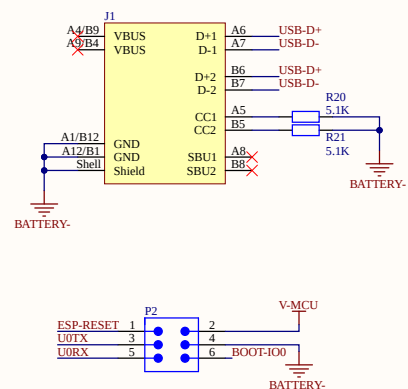
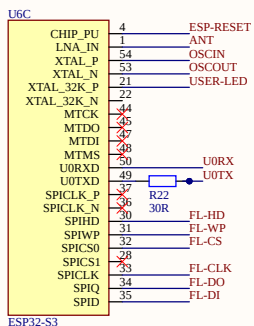
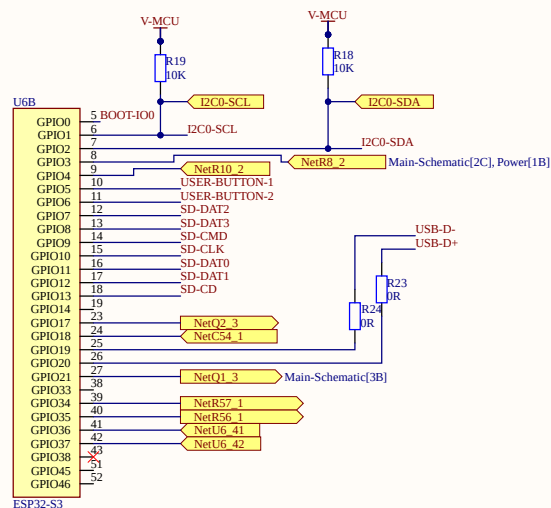


3.3V - ESP32S3

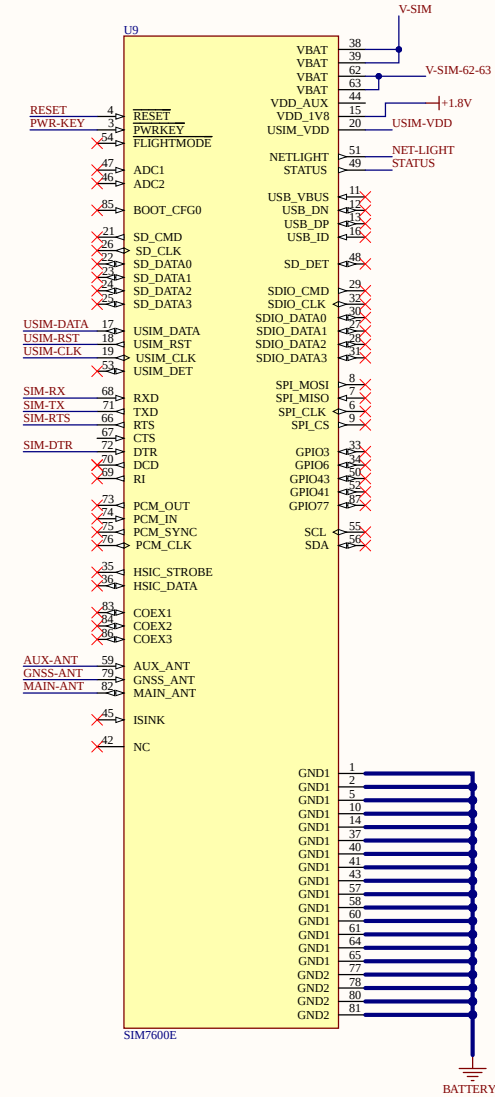


4V - SIM7600CE

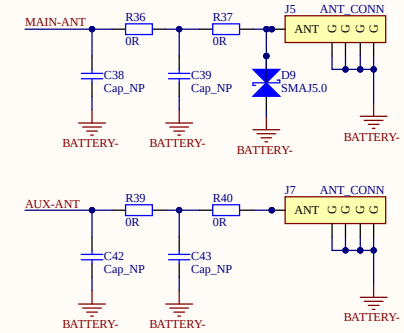
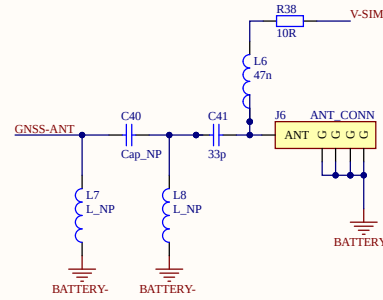




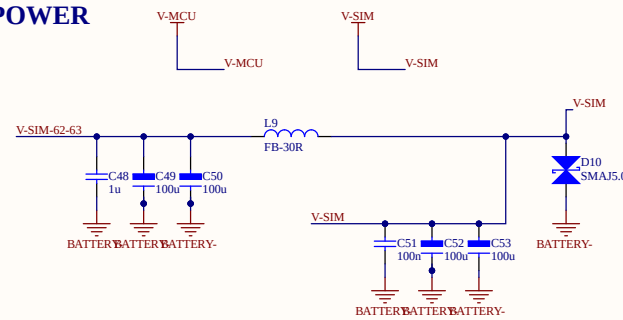
SIM7600CE



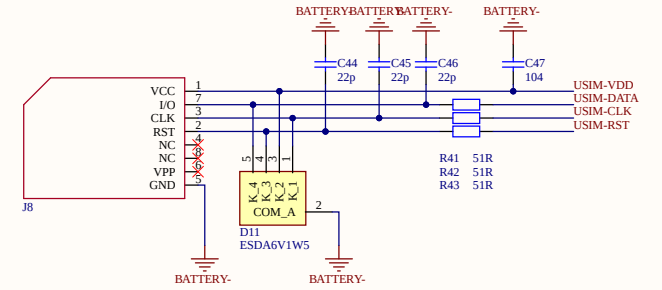
ANTENNA



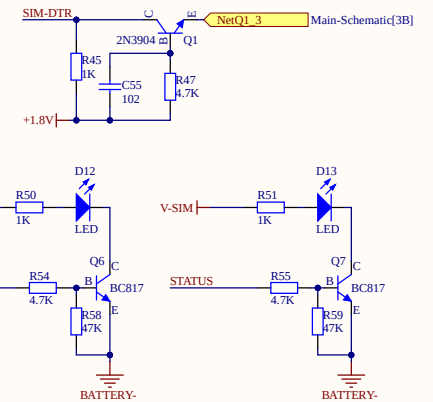
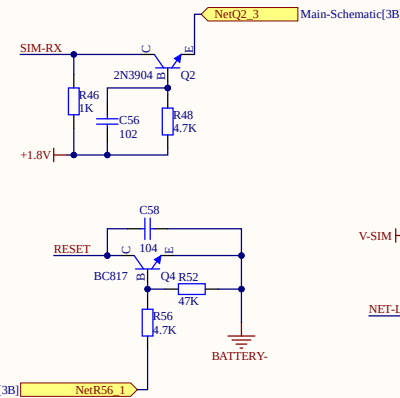
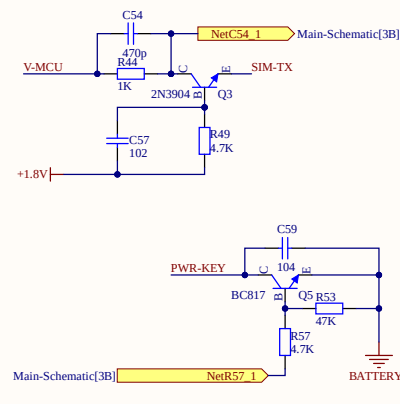
POWER

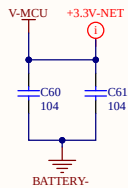
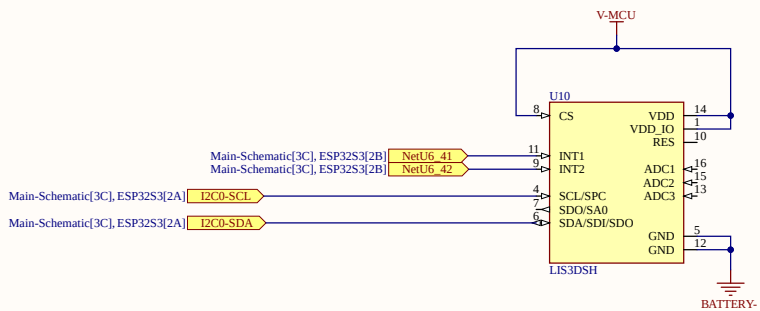


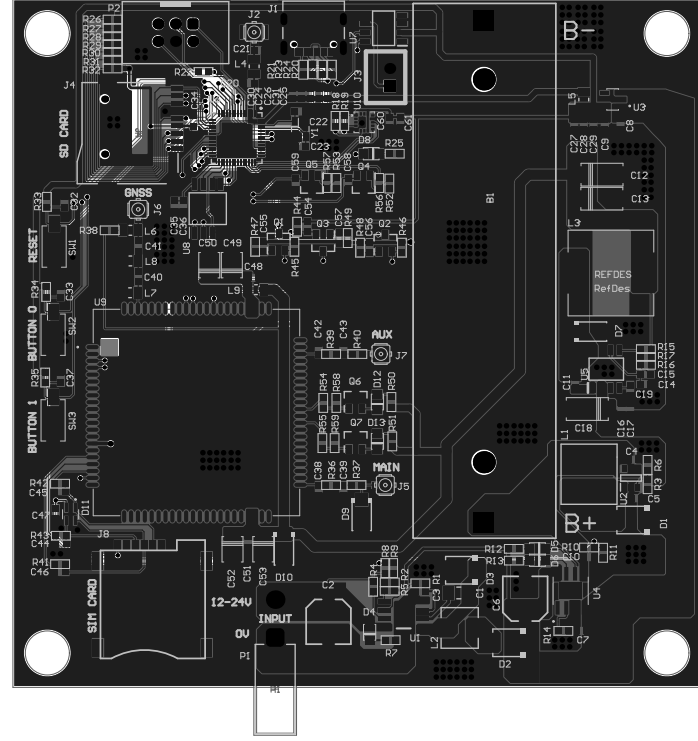
USIM



LOGIC SHIFT







Component	Description	Designator	Refdes	Library	Quantity
2100K-1C	Non-polarized capacitor	B1	18950-2C	18950-2C	1
0.1uF	Polarized aluminum capacitor	C1	Cap_1206	Cap_1NP	1
10uF	Polarized aluminum capacitor	C2	ALUMIN-CAP-6.3mm	Cap_Alu	1
1uF	Non-polarized capacitor	C3, C8, C9, C48	CAP_0603	Cap_1NP	4
22u	Non-polarized capacitor	C4, C5	CAP_0603	Cap_1NP	2
100u	Polarized aluminum capacitor	C6	ALUMIN-CAP-6.3mm	Cap_Alu	1
10u	Non-polarized capacitor	C7, C10	CAP_0603	Cap_1NP	2
0.1u	Non-polarized capacitor	C11	CAP_0603	Cap_1NP	1
47uF	Polarized tantalum capacitor	C12, C13, C18	C_Tanta_30302	Cap_Tanta	3
50p	Non-polarized capacitor	C14, C41	CAP_0603	Cap_1NP	2
1nF	Non-polarized capacitor	C15	CAP_0603	Cap_1NP	1
105	Non-polarized capacitor	C16, C19, C27, C30, C38	CAP_0603	Cap_1NP	6
104	Non-polarized capacitor	C17, C20, C28, C31, C32, C33, C34, C37, C47, C48, C50, C500, C51	CAP_0603	Cap_1NP	1
1nF	Non-polarized capacitor	C19	CAP_0603	Cap_1NP	1
10uF	Non-polarized capacitor	C20	CAP_0603	Cap_1NP	1
1.8p	Non-polarized capacitor	C21	CAP_0603	Cap_1NP	1
22p	Non-polarized capacitor	C22, C23, C44, C45, C46	CAP_0603	Cap_1NP	5
10uF	Non-polarized capacitor	C24, C36	CAP_0603	Cap_1NP	2
Cap_1NP	Non-polarized capacitor	C38, C39, C40, C42, C43	CAP_0603	Cap_1NP	5
100u	Non-polarized capacitor	C48, C50, C52, C53	C_Tanta_30302	Cap_Tanta	4
100n	Non-polarized capacitor	C51	CAP_0603	Cap_1NP	1
470p	Non-polarized capacitor	C54	CAP_0603	Cap_1NP	1
102	Non-polarized capacitor	C56, C58, C57	CAP_0603	Cap_1NP	3
3584	Schottky diode	CL, CL D3	DIODE	Diode	2
LED	Light Emitting Diode	CLD	LED-0805-Red	LED	2
LED	Light Emitting Diode	CLD, CLD2, CLD3	LED-0805-Red	LED	4
SS54	Schottky diode	CLD7	DIODE	Diode	1
58AL06	VISDIOE	CLD10	B00A	VISDIOE	1
659722W004	Integrated Circuit	U1	IC676P22001015A	EN586W04	1
659722W004	WRC4004-12.5V 2.0V 3.0V C-Rectified Horizontal Shift	U1	659722000214	659722000214	1
ANT_CONN	Antenna Connector	W, X, Y, Z	PEX	ANT_CONN	4
60007104001	WRC4004-12.5V 2.0V 3.0V C-Rectified Horizontal Shift	U1	6007104001	6007104001	1
Microelement	Microelement	U2	6007104001	6007104001	1
4.7uH	Magnetic Core Inductor	L1	IND05	Inductor Iron	1
4.7uH	Magnetic Core Inductor	L2	IND050	Inductor Iron	1
4.7uH-3A	Magnetic Core Inductor	L3	IND-304-EL27-40VDC-3.0A	Inductor Iron	1
30H	Inductor	L4	IND-0805	Inductor	1
100H	Inductor	L5	IND-0805	Inductor	1
4m	Inductor	L6	IND-0805	Inductor	1
1.8NP	Inductor	L7, L8	IND-0805	Inductor	2
RES-0805	Resistor	L9	IND-0805	Inductor	1
64800029522	64800029522	P1	64800029522	64800029522	1
61000021621	61000021621	P2	61000021621	61000021621	1
0100003	0100003	Q1, Q2, Q3	Q01-23	N-Triodes	3
0100003	0100003	Q4, Q5, Q6, Q7	Q01-23	N-Triodes	4
0100003	0100003	Q8, Q9, Q10	RES-0805	Resistor	3
0100003	0100003	Q11	RES-0805	Resistor	1
0100003	0100003	Q12	RES-0805	Resistor	1
0100003	0100003	Q13	RES-0805	Resistor	1
0100003	0100003	Q14	RES-0805	Resistor	1
0100003	0100003	Q15	RES-0805	Resistor	1
0100003	0100003	Q16	RES-0805	Resistor	1
0100003	0100003	Q17	RES-0805	Resistor	1
0100003	0100003	Q18	RES-0805	Resistor	1
0100003	0100003	Q19	RES-0805	Resistor	1
0100003	0100003	Q20	RES-0805	Resistor	1
0100003	0100003	Q21	RES-0805	Resistor	1
0100003	0100003	Q22	RES-0805	Resistor	1
0100003	0100003	Q23	RES-0805	Resistor	1
0100003	0100003	Q24	RES-0805	Resistor	1
0100003	0100003	Q25	RES-0805	Resistor	1
0100003	0100003	Q26	RES-0805	Resistor	1
0100003	0100003	Q27	RES-0805	Resistor	1
0100003	0100003	Q28	RES-0805	Resistor	1
0100003	0100003	Q29	RES-0805	Resistor	1
0100003	0100003	Q30	RES-0805	Resistor	1
0100003	0100003	Q31	RES-0805	Resistor	1
0100003	0100003	Q32	RES-0805	Resistor	1
0100003	0100003	Q33	RES-0805	Resistor	1
0100003	0100003	Q34	RES-0805	Resistor	1
0100003	0100003	Q35	RES-0805	Resistor	1
0100003	0100003	Q36	RES-0805	Resistor	1
0100003	0100003	Q37	RES-0805	Resistor	1
0100003	0100003	Q38	RES-0805	Resistor	1
0100003	0100003	Q39	RES-0805	Resistor	1
0100003	0100003	Q40	RES-0805	Resistor	1
0100003	0100003	Q41	RES-0805	Resistor	1
0100003	0100003	Q42	RES-0805	Resistor	1
0100003	0100003	Q43	RES-0805	Resistor	1
0100003	0100003	Q44	RES-0805	Resistor	1
0100003	0100003	Q45	RES-0805	Resistor	1
0100003	0100003	Q46	RES-0805	Resistor	1
0100003	0100003	Q47	RES-0805	Resistor	1
0100003	0100003	Q48	RES-0805	Resistor	1
0100003	0100003	Q49	RES-0805	Resistor	1
0100003	0100003	Q50	RES-0805	Resistor	1
0100003	0100003	Q51	RES-0805	Resistor	1
0100003	0100003	Q52	RES-0805	Resistor	1
0100003	0100003	Q53	RES-0805	Resistor	1
0100003	0100003	Q54	RES-0805	Resistor	1
0100003	0100003	Q55	RES-0805	Resistor	1
0100003	0100003	Q56	RES-0805	Resistor	1
0100003	0100003	Q57	RES-0805	Resistor	1
0100003	0100003	Q58	RES-0805	Resistor	1
0100003	0100003	Q59	RES-0805	Resistor	1
0100003	0100003	Q60	RES-0805	Resistor	1
0100003	0100003	Q61	RES-0805	Resistor	1
0100003	0100003	Q62	RES-0805	Resistor	1
0100003	0100003	Q63	RES-0805	Resistor	1
0100003	0100003	Q64	RES-0805	Resistor	1
0100003	0100003	Q65	RES-0805	Resistor	1
0100003	0100003	Q66	RES-0805	Resistor	1
0100003	0100003	Q67	RES-0805	Resistor	1
0100003	0100003	Q68	RES-0805	Resistor	1
0100003	0100003	Q69	RES-0805	Resistor	1
0100003	0100003	Q70	RES-0805	Resistor	1
0100003	0100003	Q71	RES-0805	Resistor	1
0100003	0100003	Q72	RES-0805	Resistor	1
0100003	0100003	Q73	RES-0805	Resistor	1
0100003	0100003	Q74	RES-0805	Resistor	1
0100003	0100003	Q75	RES-0805	Resistor	1
0100003	0100003	Q76	RES-0805	Resistor	1
0100003	0100003	Q77	RES-0805	Resistor	1
0100003	0100003	Q78	RES-0805	Resistor	1
0100003	0100003	Q79	RES-0805	Resistor	1
0100003	0100003	Q80	RES-0805	Resistor	1
0100003	0100003	Q81	RES-0805	Resistor	1
0100003	0100003	Q82	RES-0805	Resistor	1
0100003	0100003	Q83	RES-0805	Resistor	1
0100003	0100003	Q84	RES-0805	Resistor	1
0100003	0100003	Q85	RES-0805	Resistor	1
0100003	0100003	Q86	RES-0805	Resistor	1
0100003	0100003	Q87	RES-0805	Resistor	1
0100003	0100003	Q88	RES-0805	Resistor	1
0100003	0100003	Q89	RES-0805	Resistor	1
0100003	0100003	Q90	RES-0805	Resistor	1
0100003	0100003	Q91	RES-0805	Resistor	1
0100003	0100003	Q92	RES-0805	Resistor	1
0100003	0100003	Q93	RES-0805	Resistor	1
0100003	0100003	Q94	RES-0805	Resistor	1
0100003	0100003	Q95	RES-0805	Resistor	1
0100003	0100003	Q96	RES-0805	Resistor	1
0100003	0100003	Q97	RES-0805	Resistor	1
0100003	0100003	Q98	RES-0805	Resistor	1
0100003	0100003	Q99	RES-0805	Resistor	1
0100003	0100003	Q100	RES-0805	Resistor	1
0100003	0100003	Q101	RES-0805	Resistor	1
0100003	0100003	Q102	RES-0805	Resistor	1
0100003	0100003	Q103	RES-0805	Resistor	1
0100003	0100003	Q104	RES-0805	Resistor	1
0100003	0100003	Q105	RES-0805	Resistor	1
0100003	0100003	Q106	RES-0805	Resistor	1
0100003	0100003	Q107	RES-0805	Resistor	1
0100003	0100003	Q108	RES-0805	Resistor	1
0100003	0100003	Q109	RES-0805	Resistor	1
0100003	0100003	Q110	RES-0805	Resistor	1
0100003	0100003	Q111	RES-0805	Resistor	1
0100003	0100003	Q112	RES-0805	Resistor	1
0100003	0100003	Q113	RES-0805	Resistor	1
0100003	0100003	Q114	RES-0805	Resistor	1
0100003	0100003	Q115	RES-0805	Resistor	1
0100003	0100003	Q116	RES-0805	Resistor	1
0100003	0100003	Q117	RES-0805	Resistor	1
0100003	0100003	Q118	RES-0805	Resistor	1
0100003	0100003	Q119	RES-0805	Resistor	1
0100003	0100003	Q120	RES-0805	Resistor	1
0100003	0100003	Q121	RES-0805	Resistor	1
0100003	0100003	Q122	RES-0805	Resistor	1
0100003	0100003	Q123	RES-0805	Resistor	1
0100003	0100003	Q124	RES-0805	Resistor	1
0100003	0100003	Q125	RES-0805	Resistor	1
0100003	0100003	Q126	RES-0805	Resistor	1
0100003	0100003	Q127	RES-0805	Resistor	1
0100003	0100003	Q128	RES-0805	Resistor	1
0100003	0100003	Q129	RES-0805	Resistor	1
0100003	0100003	Q130	RES-0805	Resistor	1
0100003	0100003	Q131	RES-0805	Resistor	1
0100003	0100003	Q132	RES-0805	Resistor	1
0100003	0100003	Q133	RES-0805	Resistor	1
0100003	0100003	Q134	RES-0805	Resistor	1
0100003	0100003	Q135	RES-0805	Resistor	1
0100003	0100003	Q136	RES-0805	Resistor	1
0100003	0100003	Q137	RES-0805	Resistor	1
0100003	0100003	Q138	RES-0805	Resistor	1
0100003	0100003	Q139	RES-0805	Resistor	1
0100003	0100003	Q140	RES-0805	Resistor	1
0100003	0100003	Q141	RES-0805	Resistor	1
0100003	0100003	Q142	RES-0805	Resistor	1
0100003	0100003	Q143	RES-0805	Resistor	1
0100003	0100003	Q144	RES-0805	Resistor	1
0100003	0100003	Q145	RES-0805	Resistor	1
0100003	0100003	Q146	RES-0805	Resistor	1
0100003	0100003	Q147	RES-0805	Resistor	1
0100003	0100003	Q148	RES-0805	Resistor	1
0100003	0100003	Q149	RES-0805	Resistor	1
0100003	0100003	Q150	RES-0805	Resistor	1
0100003	0100003	Q151	RES-0805	Resistor	1
0100003	0100003	Q152	RES-0805	Resistor	1
0100003	0100003	Q153	RES-0805	Resistor	1
0100003	0100003	Q154	RES-0805	Resistor	1
0100003	0100003	Q155	RES-0805	Resistor	1
0100003	0100003	Q156	RES-0805	Resistor	1
0100003	0100003	Q157	RES-0805	Resistor	1
0100003	0100003	Q158	RES-0805	Resistor	1
0100003	0100003	Q159	RES-0805	Resistor	1
0100003	0100003	Q160	RES-0805	Resistor	1
0100003	0100003	Q161	RES-0805	Resistor	1
0100003	0100003	Q162	RES-0805	Resistor	1
0100003	0100003	Q163	RES-0805	Resistor	1
0100003	0100003	Q164	RES-0805	Resistor	1
0100003	0100003	Q165	RES-0805	Resistor	1
0100003	0100003	Q166	RES-0805	Resistor	1
0100003	0100003	Q167	RES-0805</		